Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: R & S Dairy

Physical address of dairy:

Number and Street	City	County	Zip Code
24065 E Arthur RD	Escalon	San Joaquin	95320

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1970

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X229-X070-XX10-XXXX

B. OPERATORS

Silva, Johnny

Operator name: Silva, Johnny

Telephone no.: Landline: ; Cellular: (209) 652-4989

Mailing Address Number and Street	City	State	Zip Code
23243 Lone Tree RD	Escalon	CA	95320

This operator is responsible for paying permit fees.

C. OWNERS

Silva, Johnny

Legal owner name: Silva, Johnny

Telephone no.: Landline: ; Cellular: (209) 652-4989

Mailing Address Number and Street	City	State	Zip Code
23243 Lone Tree RD	Escalon	CA	95320

This owner is responsible for paying permit fees.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	295	200	0	0
Number under roof	600	0	0	0	0	0
Maximum number	600	0	295	200	0	0
Average number	600	0	295	200	0	0
Avg live weight (lbs)	1,400	0	1,000	700		

Predominant milk cow breed: Holstein

Average milk production: 78 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 20,935.08 *tons per reporting period*

Total nitrogen from manure: 255,568.97 *lbs per reporting period*

After ammonia losses (30% loss applied): 178,898.28 *lbs per reporting period*

Total phosphorus from manure: 43,370.70 *lbs per reporting period*

Total potassium from manure: 118,059.14 *lbs per reporting period*

Total salt from manure: 282,510.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 14,364,000 *gallons*

Total nitrogen generated: 45,110.17 *lbs*

Total phosphorus generated: 6,670.16 *lbs*

Total potassium generated: 62,139.45 *lbs*

Total salt generated: 433,987.23 *lbs*

	14,364,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	14,364,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
05/16/2023	Solid commercial fertilizer *4-0-0*	20.80 *ton*	Dry-weight	0.1	4.000000	0.000000	0.000000	0.000000
06/14/2023	Solid commercial fertilizer *UN32*	5.64 *ton*	Dry-weight	0.1	32.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	5,268.33	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	5,268.33	0.00	0.00	0.00

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Field 1	23	23	2	process wastewater	X229-X070-XX10-XXXX
Field 2	27	27	2	process wastewater	X229-X070-XX10-XXXX
Field 3	17	17	2	process wastewater	X229-X070-XX10-XXXX
Field 4	17	17	1	none	X225-X020-XX40-XXXX
Field 5	94	94	2	manure	X225-X020-XX29-XXXX X225-X020-XX30-XXXX X225-X020-XX31-XXXX X225-X020-XX40-XXXX
Field 6	47	47	2	process wastewater	X229-X070-XX01-XXXX X229-X070-XX04-XXXX
Totals for areas that were used for application	208	208	10		
Totals for areas that were not used for application	17	17	1		
Land application area totals	225	225	11		

B. CROPS AND HARVESTS

Field 1

Field name: Field 1

10/22/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 23 Plant date: 10/22/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/01/2023	387.20 *ton*	As-is		62.0	3,350.00	1,030.00	6,360.00		6.27

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.83	112.79	34.68	214.14	802.21

Field 1

05/17/2023: Corn, silage

Crop: Corn, silage Acres planted: 23 Plant date: 05/17/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/07/2023	656.40 *ton*	As-is		70.0	3,420.00	490.00	4,120.00		5.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	28.54	195.21	27.97	235.16	976.04

Field 2

Field name: Field 2

10/23/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 27 Plant date: 10/23/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/01/2023	445.50 *ton*	As-is		63.6	4,030.00	1,230.00	9,330.00		8.25

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.50	132.99	40.59	307.89	990.99

05/30/2023: Corn, silage

Crop: Corn, silage Acres planted: 27 Plant date: 05/30/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/23/2023	761.40 *ton*	As-is		67.6	3,400.00	1,210.00	5,470.00		7.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	28.20	191.76	68.24	308.51	1,315.70

Field 3

Field name: Field 3

10/23/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 17 Plant date: 10/23/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/30/2023	283.90 *ton*	As-is		71.0	4,940.00	1,080.00	9,220.00		10.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.70	165.00	36.07	307.95	978.29

05/30/2023: Corn, silage

Crop: Corn, silage Acres planted: 17 Plant date: 05/30/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/23/2023	482.80 *ton*	As-is		67.6	3,400.00	1,210.00	5,470.00		7.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	28.40	193.12	68.73	310.70	1,325.03

Field 4

Field name: Field 4

01/01/2023: Pasture

Crop: Pasture Acres planted: 17 Plant date: 01/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
12/31/2023	255.00 *ton*	As-is		70.0	7,260.00	1,950.00	14,400.00		41.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	276.00	39.00	283.50	0.00
Total actual harvest content	15.00	217.80	58.50	432.00	3,690.00

Field 5

Field name: Field 5

10/24/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 94 Plant date: 10/24/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/02/2023	1,545.20 *ton*	As-is		64.2	3,830.00	890.00	4,790.00		8.47

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.44	125.92	29.26	157.48	996.90

05/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 94 Plant date: 05/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/03/2023	2,679.00 *ton*	As-is		68.4	3,980.00	650.00	3,480.00		6.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	28.50	226.86	37.05	198.36	1,080.72

Field 6

Field name: Field 6

10/25/2002: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 47 Plant date: 10/25/2002

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/02/2023	747.30 *ton*	As-is		67.1	4,640.00	910.00	5,660.00		9.05

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.90	147.55	28.94	179.99	946.83

Field 6

05/16/2023: Corn, silage

Crop: Corn, silage Acres planted: 47 Plant date: 05/16/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/03/2023	1,301.90 *ton*	As-is		68.4	3,980.00	650.00	3,480.00		6.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	27.70	220.49	36.01	192.79	1,050.38

NUTRIENT BUDGET

A. LAND APPLICATIONS

Field 1 - 10/22/2022: Oats, silage-soft dough

Field name: Field 1

Crop: Oats, silage-soft dough — Plant date: 10/22/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	46.62	4.20	108.43	975.73	399,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	0.00	2,730,000.00 *gal*
Application event totals		46.62	4.20	108.43	975.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	98.67	16.99	153.52	1,145.32	741,000.00 *gal*
Application event totals		98.67	16.99	153.52	1,145.32	

Field 1 - 05/17/2023: Corn, silage

Field name: Field 1

Crop: Corn, silage — Plant date: 05/17/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	66.08	12.22	75.38	359.85	427,500.00 *gal*
Canal	Surface water	0.00	0.00	0.00	61.55	2,925,000.00 *gal*
Application event totals		66.08	12.22	75.38	421.40	

Field 1 - 05/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/17/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	65.66	3,120,000.00 *gal*
Application event totals		0.00	0.00	0.00	65.66	

07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	54.72	6.56	57.47	367.71	399,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	57.45	2,730,000.00 *gal*
Application event totals		54.72	6.56	57.47	425.16	

07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	54.72	6.56	57.47	367.71	399,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	57.45	2,730,000.00 *gal*
Application event totals		54.72	6.56	57.47	425.16	

07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	65.66	3,120,000.00 *gal*
Application event totals		0.00	0.00	0.00	65.66	

Field 1 - 05/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	54.72	6.56	57.47	367.71	399,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	57.45	2,730,000.00 *gal*
Application event totals		54.72	6.56	57.47	425.16	

Field 2 - 10/23/2022: Oats, silage-soft dough

Field name: Field 2

Crop: Oats, silage-soft dough

Plant date: 10/23/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	42.55	3.83	98.96	890.55	427,500.00 *gal*
Canal	Surface water	0.00	0.00	0.00	0.00	2,925,000.00 *gal*
Application event totals		42.55	3.83	98.96	890.55	

01/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	64.66	11.13	100.59	750.49	570,000.00 *gal*
Application event totals		64.66	11.13	100.59	750.49	

02/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	58.19	10.02	90.53	675.44	513,000.00 *gal*
Application event totals		58.19	10.02	90.53	675.44	

Field 2 - 05/30/2023: Corn, silage

Field 2 - 05/30/2023: Corn, silage

Field name: Field 2

Crop: Corn, silage

Plant date: 05/30/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	63.79	11.80	72.78	347.41	484,500.00 *gal*
Canal	Surface water	0.00	0.00	0.00	59.43	3,315,000.00 *gal*
Application event totals		63.79	11.80	72.78	406.84	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/30/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	66.42	3,705,000.00 *gal*
Application event totals		0.00	0.00	0.00	66.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	53.27	6.38	55.95	357.98	456,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	55.93	3,120,000.00 *gal*
Application event totals		53.27	6.38	55.95	413.91	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	53.27	6.38	55.95	357.98	456,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	55.93	3,120,000.00 *gal*
Application event totals		53.27	6.38	55.95	413.91	

Field 2 - 05/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	49.94	5.99	52.46	335.61	427,500.00 *gal*
Canal	Surface water	0.00	0.00	0.00	52.43	2,925,000.00 *gal*
Application event totals		49.94	5.99	52.46	388.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	62.92	3,510,000.00 *gal*
Application event totals		0.00	0.00	0.00	62.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	55.93	3,120,000.00 *gal*
Application event totals		0.00	0.00	0.00	55.93	

Field 3 - 10/23/2022: Oats, silage-soft dough

Field name: Field 3

Crop: Oats, silage-soft dough

Plant date: 10/23/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/02/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	45.05	4.06	104.79	942.94	285,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	0.00	1,950,000.00 *gal*
Application event totals		45.05	4.06	104.79	942.94	

Field 3 - 10/23/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	71.88	12.38	111.84	834.37	399,000.00 *gal*
Application event totals		71.88	12.38	111.84	834.37	

03/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	71.88	12.38	111.84	834.37	399,000.00 *gal*
Application event totals		71.88	12.38	111.84	834.37	

Field 3 - 05/30/2023: Corn, silage

Field name: Field 3

Crop: Corn, silage

Plant date: 05/30/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	65.56	12.13	74.79	357.03	313,500.00 *gal*
Canal	Surface water	0.00	0.00	0.00	61.07	2,145,000.00 *gal*
Application event totals		65.56	12.13	74.79	418.10	

05/30/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	66.62	2,340,000.00 *gal*
Application event totals		0.00	0.00	0.00	66.62	

Field 3 - 05/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	52.88	6.34	55.54	355.35	285,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	55.52	1,950,000.00 *gal*
Application event totals		52.88	6.34	55.54	410.87	

07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	52.88	6.34	55.54	355.35	285,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	55.52	1,950,000.00 *gal*
Application event totals		52.88	6.34	55.54	410.87	

08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	52.88	6.34	55.54	355.35	285,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	55.52	1,950,000.00 *gal*
Application event totals		52.88	6.34	55.54	410.87	

08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	61.07	2,145,000.00 *gal*
Application event totals		0.00	0.00	0.00	61.07	

09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	55.52	1,950,000.00 *gal*
Application event totals		0.00	0.00	0.00	55.52	

Field 4 - 01/01/2023: Pasture

Field name: Field 4

Crop: Pasture

Plant date: 01/01/2023

Field 4 - 01/01/2023: Pasture

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	44.42	1,560,000.00 *gal*
Application event totals		0.00	0.00	0.00	44.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	44.42	1,560,000.00 *gal*
Application event totals		0.00	0.00	0.00	44.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	44.42	1,560,000.00 *gal*
Application event totals		0.00	0.00	0.00	44.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	44.42	1,560,000.00 *gal*
Application event totals		0.00	0.00	0.00	44.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	47.19	1,657,500.00 *gal*
Application event totals		0.00	0.00	0.00	47.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	47.19	1,657,500.00 *gal*
Application event totals		0.00	0.00	0.00	47.19	

Field 4 - 01/01/2023: Pasture

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	47.19	1,657,500.00 *gal*
Application event totals		0.00	0.00	0.00	47.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	47.19	1,657,500.00 *gal*
Application event totals		0.00	0.00	0.00	47.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	44.42	1,560,000.00 *gal*
Application event totals		0.00	0.00	0.00	44.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	44.42	1,560,000.00 *gal*
Application event totals		0.00	0.00	0.00	44.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	44.42	1,560,000.00 *gal*
Application event totals		0.00	0.00	0.00	44.42	

Field 5 - 10/24/2022: Oats, silage-soft dough

Field name: Field 5

Crop: Oats, silage-soft dough

Plant date: 10/24/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 5 - 10/24/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/30/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	186.26	102.40	401.66	4,051.90	900.00 *ton*
Application event totals		186.26	102.40	401.66	4,051.90	

10/04/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	0.00	11,700,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Field 5 - 05/15/2023: Corn, silage

Field name: Field 5

Crop: Corn, silage

Plant date: 05/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/03/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	218.39	170.74	829.86	34,147.49	2,150.00 *ton*
Application event totals		218.39	170.74	829.86	34,147.49	

05/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	68.28	13,260,000.00 *gal*
Application event totals		0.00	0.00	0.00	68.28	

05/15/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

Field 5 - 05/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/14/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	38.40	0.00	0.00	0.00	
Application event totals		38.40	0.00	0.00	0.00	

06/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	64.26	12,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	64.26	

06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	62.25	12,090,000.00 *gal*
Application event totals		0.00	0.00	0.00	62.25	

07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	64.26	12,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	64.26	

07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	64.26	12,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	64.26	

08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	62.25	12,090,000.00 *gal*
Application event totals		0.00	0.00	0.00	62.25	

Field 6 - 10/25/2002: Oats, silage-soft dough

Field 6 - 10/25/2002: Oats, silage-soft dough

Field name: Field 6

Crop: Oats, silage-soft dough

Plant date: 10/25/2002

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/02/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	43.99	3.96	102.33	920.87	769,500.00 *gal*
Canal	Surface water	0.00	0.00	0.00	0.00	5,265,000.00 *gal*
Application event totals		43.99	3.96	102.33	920.87	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	66.86	11.51	104.02	776.04	1,026,000.00 *gal*
Application event totals		66.86	11.51	104.02	776.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	74.28	12.79	115.58	862.27	1,140,000.00 *gal*
Application event totals		74.28	12.79	115.58	862.27	

Field 6 - 05/16/2023: Corn, silage

Field name: Field 6

Crop: Corn, silage

Plant date: 05/16/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	60.36	11.16	68.86	328.71	798,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	56.23	5,460,000.00 *gal*
Application event totals		60.36	11.16	68.86	384.94	

Field 6 - 05/16/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/16/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

06/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	64.26	6,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	64.26	

06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	58.20	10.77	66.40	316.98	769,500.00 *gal*
Canal	Surface water	0.00	0.00	0.00	54.22	5,265,000.00 *gal*
Application event totals		58.20	10.77	66.40	371.19	

07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	51.65	6.19	54.24	347.03	769,500.00 *gal*
Canal	Surface water	0.00	0.00	0.00	54.22	5,265,000.00 *gal*
Application event totals		51.65	6.19	54.24	401.25	

07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	64.26	6,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	64.26	

Field 6 - 05/16/2023: Corn, silage

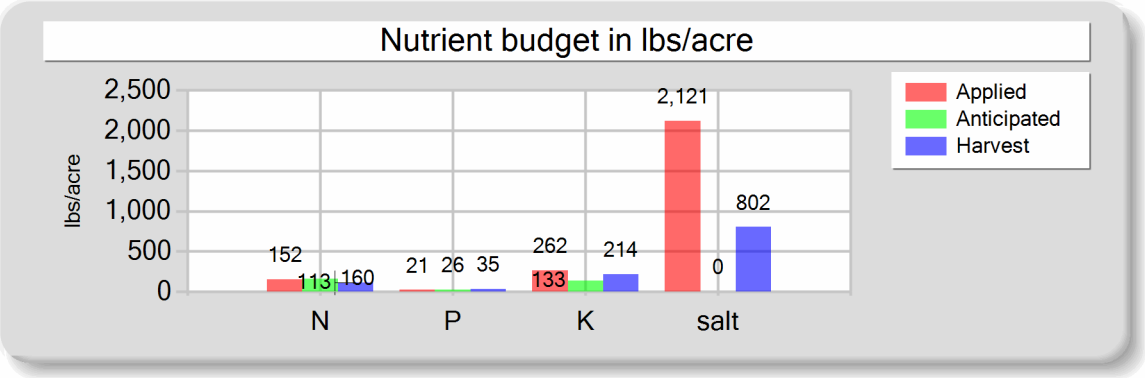
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	49.73	5.96	52.23	334.18	741,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	52.21	5,070,000.00 *gal*
Application event totals		49.73	5.96	52.23	386.39	

B. NUTRIENT BUDGET

Field 1 - 10/22/2022: Oats, silage-soft dough

Field name: Field 1 Crop: Oats, silage-soft dough Plant date: 10/22/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	145.28	21.19	261.95	2,121.05
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	152.28	21.19	261.95	2,121.05
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	112.79	34.68	214.14	802.21
Nutrient balance	39.49	-13.49	47.81	1,318.84
Applied to removed ratio	1.35	0.61	1.22	2.64

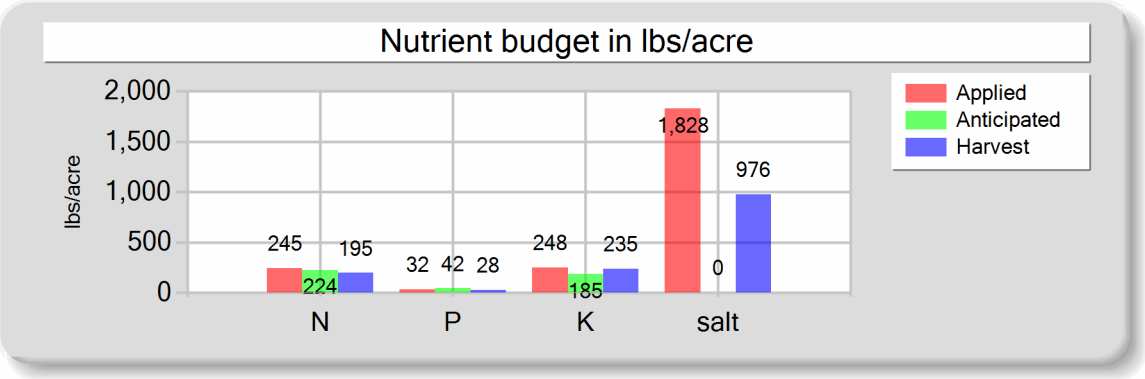
Fresh water applied	
2,730,000.00	*gallons*
100.54	*acre-inches*
4.37	*inches/acre*

Process wastewater applied	
1,140,000.00	*gallons*
41.98	*acre-inches*
1.83	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 1 - 05/17/2023: Corn, silage

Field name: Field 1 Crop: Corn, silage Plant date: 05/17/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	230.24	31.90	247.80	1,462.98
Fresh water	0.00	0.00	0.00	365.22
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	245.24	31.90	247.80	1,828.20
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	195.21	27.97	235.16	976.04
Nutrient balance	50.03	3.93	12.64	852.16
Applied to removed ratio	1.26	1.14	1.05	1.87

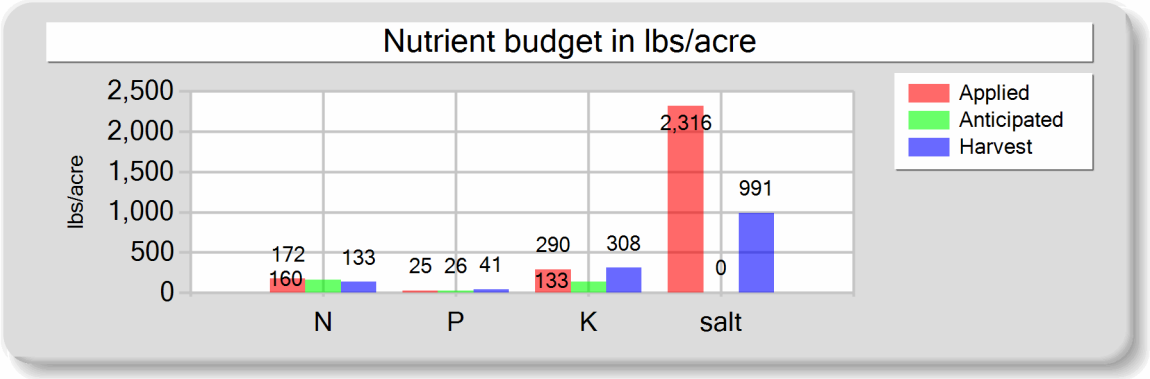
Fresh water applied	
17,355,000.00	*gallons*
639.13	*acre-inches*
27.79	*inches/acre*

Process wastewater applied	
1,624,500.00	*gallons*
59.82	*acre-inches*
2.60	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 10/23/2022: Oats, silage-soft dough

Field name: Field 2 Crop: Oats, silage-soft dough Plant date: 10/23/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	165.39	24.99	290.09	2,316.49
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	172.39	24.99	290.09	2,316.49
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	132.99	40.59	307.89	990.99
Nutrient balance	39.40	-15.60	-17.80	1,325.50
Applied to removed ratio	1.30	0.62	0.94	2.34

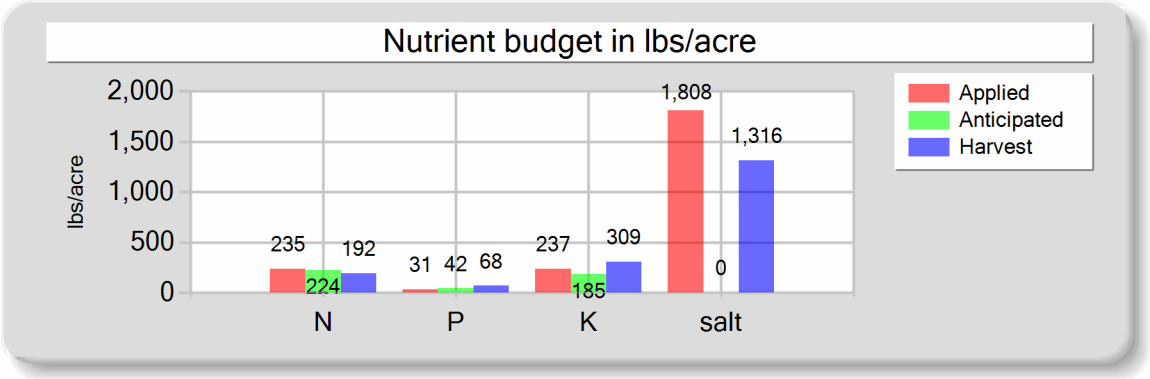
Fresh water applied	
2,925,000.00	*gallons*
107.72	*acre-inches*
3.99	*inches/acre*

Process wastewater applied	
1,510,500.00	*gallons*
55.63	*acre-inches*
2.06	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 05/30/2023: Corn, silage

Field name: Field 2 Crop: Corn, silage Plant date: 05/30/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	220.29	30.55	237.14	1,398.98
Fresh water	0.00	0.00	0.00	408.99
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	235.29	30.55	237.14	1,807.97
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	191.76	68.24	308.51	1,315.70
Nutrient balance	43.53	-37.69	-71.37	492.27
Applied to removed ratio	1.23	0.45	0.77	1.37

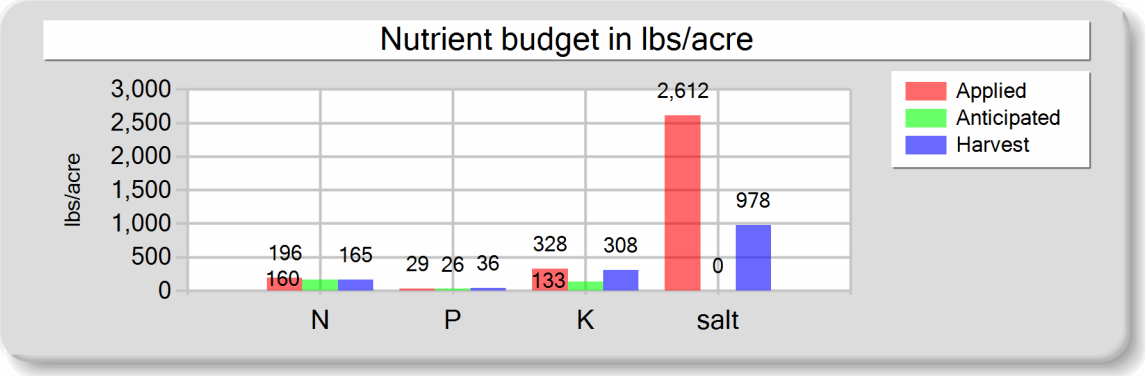
Fresh water applied	
22,815,000.00	*gallons*
840.20	*acre-inches*
31.12	*inches/acre*

Process wastewater applied	
1,824,000.00	*gallons*
67.17	*acre-inches*
2.49	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 10/23/2022: Oats, silage-soft dough

Field name: Field 3 Crop: Oats, silage-soft dough Plant date: 10/23/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	188.81	28.81	328.46	2,611.68
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	195.81	28.81	328.46	2,611.68
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	165.00	36.07	307.95	978.29
Nutrient balance	30.82	-7.26	20.51	1,633.39
Applied to removed ratio	1.19	0.80	1.07	2.67

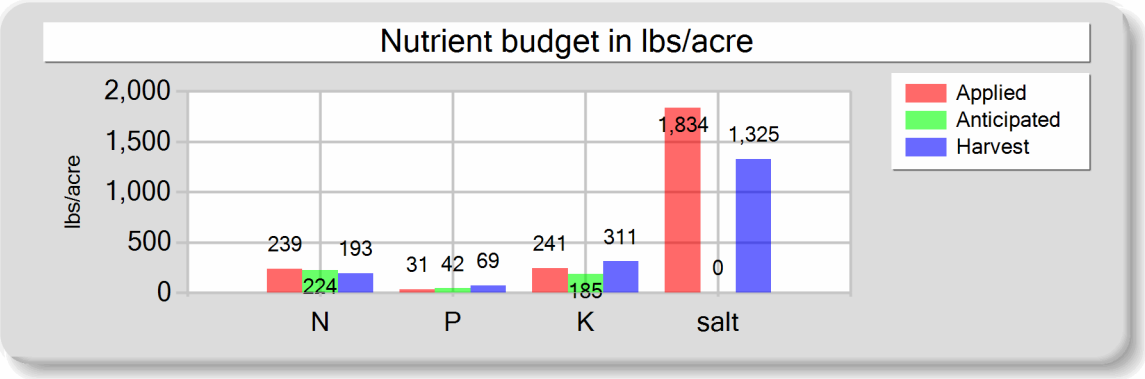
Fresh water applied	
1,950,000.00	*gallons*
71.81	*acre-inches*
4.22	*inches/acre*

Process wastewater applied	
1,083,000.00	*gallons*
39.88	*acre-inches*
2.35	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 05/30/2023: Corn, silage

Field name: Field 3 Crop: Corn, silage Plant date: 05/30/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	224.21	31.14	241.41	1,423.08
Fresh water	0.00	0.00	0.00	410.84
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	239.21	31.14	241.41	1,833.92
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	193.12	68.73	310.70	1,325.03
Nutrient balance	46.09	-37.59	-69.28	508.89
Applied to removed ratio	1.24	0.45	0.78	1.38

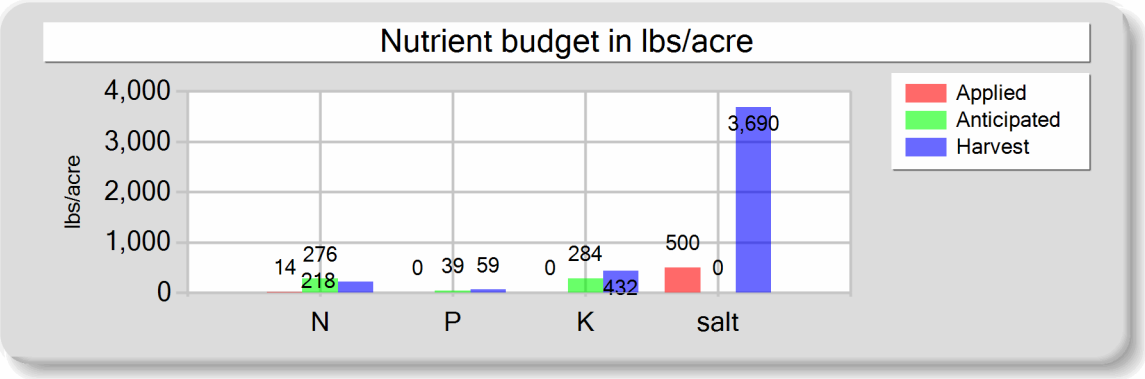
Fresh water applied	
14,430,000.00	*gallons*
531.41	*acre-inches*
31.26	*inches/acre*

Process wastewater applied	
1,168,500.00	*gallons*
43.03	*acre-inches*
2.53	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 4 - 01/01/2023: Pasture

Field name: Field 4 Crop: Pasture Plant date: 01/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	499.67
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	14.00	0.00	0.00	499.67
Anticipated crop nutrient removal	276.00	39.00	283.50	0.00
Actual crop nutrient removal	217.80	58.50	432.00	3,690.00
Nutrient balance	-203.80	-58.50	-432.00	-3,190.33
Applied to removed ratio	0.06	0.00	0.00	0.14

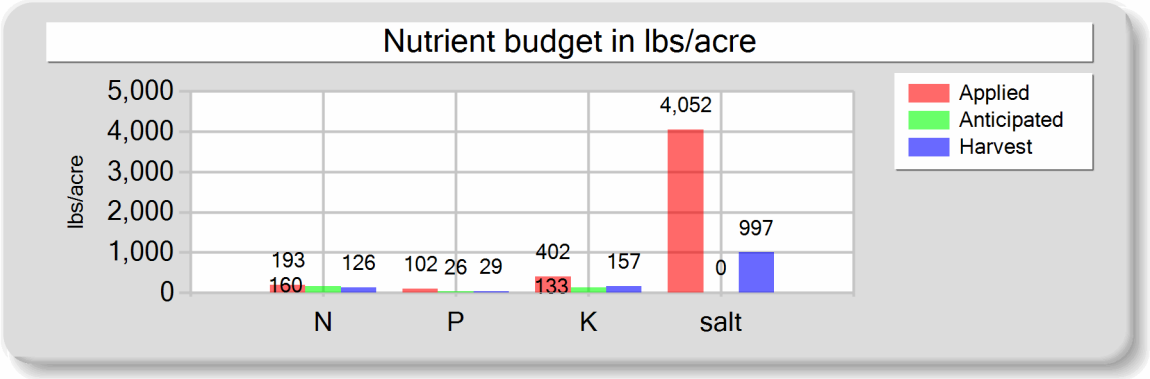
Fresh water applied	
17,550,000.00	*gallons*
646.31	*acre-inches*
38.02	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 5 - 10/24/2022: Oats, silage-soft dough

Field name: Field 5 Crop: Oats, silage-soft dough Plant date: 10/24/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	186.26	102.40	401.66	4,051.90
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	193.26	102.40	401.66	4,051.90
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	125.92	29.26	157.48	996.90
Nutrient balance	67.35	73.14	244.18	3,054.99
Applied to removed ratio	1.53	3.50	2.55	4.06

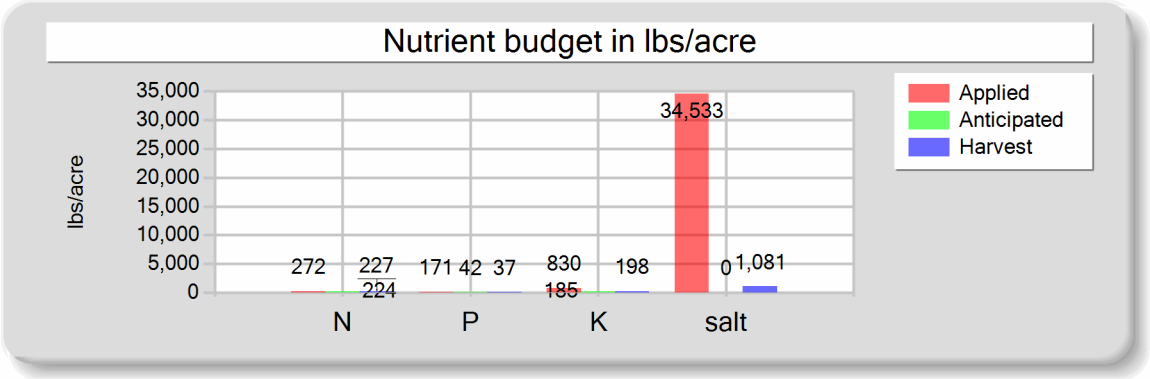
Fresh water applied	
11,700,000.00	*gallons*
430.87	*acre-inches*
4.58	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 5 - 05/15/2023: Corn, silage

Field name: Field 5 Crop: Corn, silage Plant date: 05/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	46.40	0.00	0.00	0.00
Dry manure	218.39	170.74	829.86	34,147.49
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	385.56
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	271.79	170.74	829.86	34,533.05
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	226.86	37.05	198.36	1,080.72
Nutrient balance	44.93	133.69	631.50	33,452.33
Applied to removed ratio	1.20	4.61	4.18	31.95

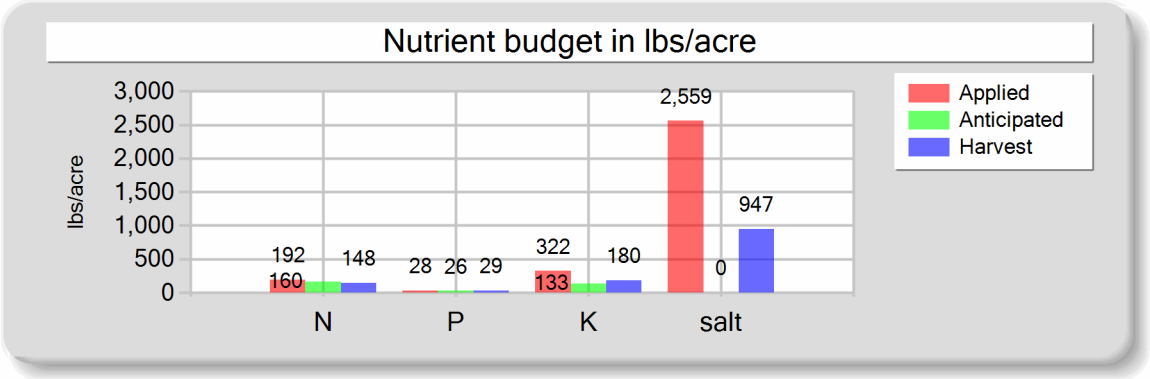
Fresh water applied	
74,880,000.00	*gallons*
2,757.58	*acre-inches*
29.34	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 6 - 10/25/2002: Oats, silage-soft dough

Field name: Field 6 Crop: Oats, silage-soft dough Plant date: 10/25/2002



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	185.13	28.27	321.93	2,559.18
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	192.13	28.27	321.93	2,559.18
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	147.55	28.94	179.99	946.83
Nutrient balance	44.58	-0.67	141.94	1,612.35
Applied to removed ratio	1.30	0.98	1.79	2.70

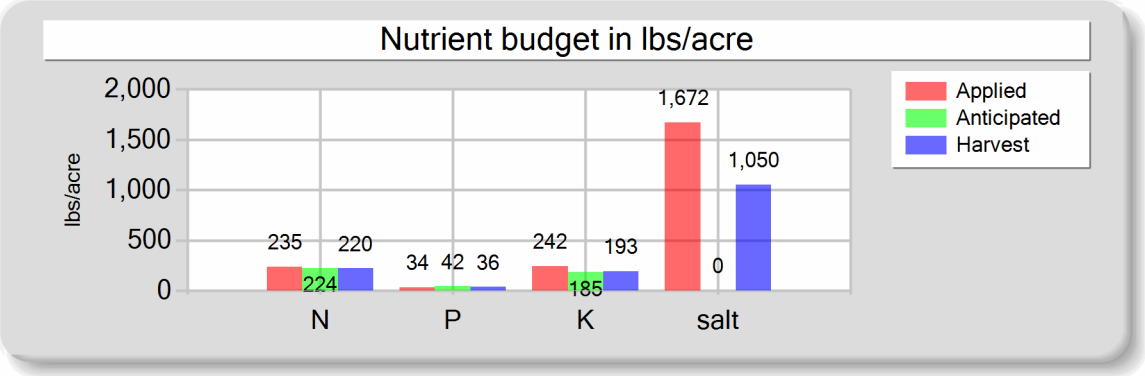
Fresh water applied	
5,265,000.00	*gallons*
193.89	*acre-inches*
4.13	*inches/acre*

Process wastewater applied	
2,935,500.00	*gallons*
108.10	*acre-inches*
2.30	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 6 - 05/16/2023: Corn, silage

Field name: Field 6 Crop: Corn, silage Plant date: 05/16/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	219.94	34.08	241.73	1,326.90
Fresh water	0.00	0.00	0.00	345.40
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	234.94	34.08	241.73	1,672.30
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	220.49	36.01	192.79	1,050.38
Nutrient balance	14.45	-1.93	48.94	621.92
Applied to removed ratio	1.07	0.95	1.25	1.59

Fresh water applied	
33,540,000.00	*gallons*
1,235.16	*acre-inches*
26.28	*inches/acre*

Process wastewater applied	
3,078,000.00	*gallons*
113.35	*acre-inches*
2.41	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Corral Solids

Sample and source description: Corral Solids

Sample date: 09/12/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 53.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,100.00	11,600.00	45,500.00	23,900.00	11,300.00	5,160.00	6,100.00	1,660.00		45.90
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Corral Manure

Sample and source description: Corral Manure

Sample date: 04/17/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 13.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,500.00	4,300.00	20,900.00	10,300.00	5,400.00	3,000.00	3,000.00	1,370.00		86.00
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Compost

Sample and source description: Compost

Sample date: 10/09/2023 Material type: Compost Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 4.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,800.00	5,000.00	16,700.00	0.00	0.00	0.00	0.00	0.00		58.70
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 10/09/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 51.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,100.00	9,680.00	6,570.00	0.00	0.00	0.00	0.00	0.00		66.60
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

B. PROCESS WASTEWATER ANALYSES

WWQ4

Sample and source description: WWQ4

Sample date: 10/24/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	322.00	219.00	0.00	0.00	29.00	749.00								5,320.00	6,740
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ1

Sample and source description: WWQ1

Sample date: 02/21/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.60

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	367.00	232.00	0.00	0.00	63.20	571.00	197.00	90.60	127.00	2,260.00	0.00	274.00	189.00	4,810.00	4,260
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ2

Sample and source description: WWQ2

Sample date: 04/10/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.50

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	426.00	118.00	0.00	0.00	78.80	486.00								3,020.00	2,320
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ3

Sample and source description: WWQ3

Sample date: 07/26/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.40

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	378.00	249.00	0.00	0.00	45.30	397.00								3,940.00	2,540
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ4

Sample and source description: WWQ4

Sample date: 10/30/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.30

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	302.00	199.00	0.00	0.00	69.50	352.00								3,590.00	2,360
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

Canal

Canal

Canal

Sample description: Canal

Sample date: 08/11/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								0.00	0
DL	0.50	0.50	0.50								1.00	10

SSJID

Sample description: SSJID

Sample date: 08/17/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								89.20	58
DL	1.00	0.50	0.50								10.00	10

D. SOIL ANALYSES

Field 1

Soil

Sample and source description: Soil

Sample date: 12/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			63.00				
DL			4.00				

Field 2

Field 2

Soil

Sample and source description: Soil

Sample date: 12/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			86.00				
DL			4.00				

Field 3

Soil

Sample and source description: Soil

Sample date: 12/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			63.00				
DL			4.00				

Field 4

Soil

Sample and source description: Soil

Sample date: 12/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			86.00				
DL			4.00				

Field 5

Field 5

Soil

Sample and source description: Soil

Sample date: 12/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			102.00				
DL			4.00				

Field 6

Soil

Sample and source description: Soil

Sample date: 12/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			88.00				
DL			4.00				

E. PLANT TISSUE ANALYSES

Field 1 - 10/22/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/01/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 62.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,350.00	1,030.00	6,360.00		6.27
DL	100.00	100.00	100.00		0.01

Field 1 - 05/17/2023: Corn, silage

Field 1 - 05/17/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,420.00	490.00	4,120.00		5.70
DL	100.00	100.00	100.00		0.01

Field 2 - 10/23/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/01/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,030.00	1,230.00	9,330.00		8.25
DL	100.00	100.00	100.00		0.01

Field 2 - 05/30/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,400.00	1,210.00	5,470.00		7.20
DL	100.00	100.00	100.00		0.01

Field 3 - 10/23/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/01/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,940.00	1,080.00	9,220.00		10.10
DL	100.00	100.00	100.00		0.01

Field 3 - 05/30/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,400.00	1,210.00	5,470.00		7.20
DL	100.00	100.00	100.00		0.01

Field 4 - 01/01/2023: Pasture

Pasture

Sample and source description: Pasture

Sample date: 05/06/2022 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 79.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,910.00	1,320.00	9,740.00		41.00
DL	100.00	100.00	100.00		0.01

Field 5 - 10/24/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 09/08/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,830.00	890.00	4,790.00		8.47
DL	100.00	100.00	100.00		0.01

Field 5 - 05/15/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,980.00	650.00	3,480.00		6.00
DL	100.00	100.00	100.00		0.01

Field 6 - 10/25/2002: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 09/08/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,640.00	910.00	5,660.00		9.05
DL	100.00	100.00	100.00		0.01

Field 6 - 05/16/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,980.00	650.00	3,480.00		6.00
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

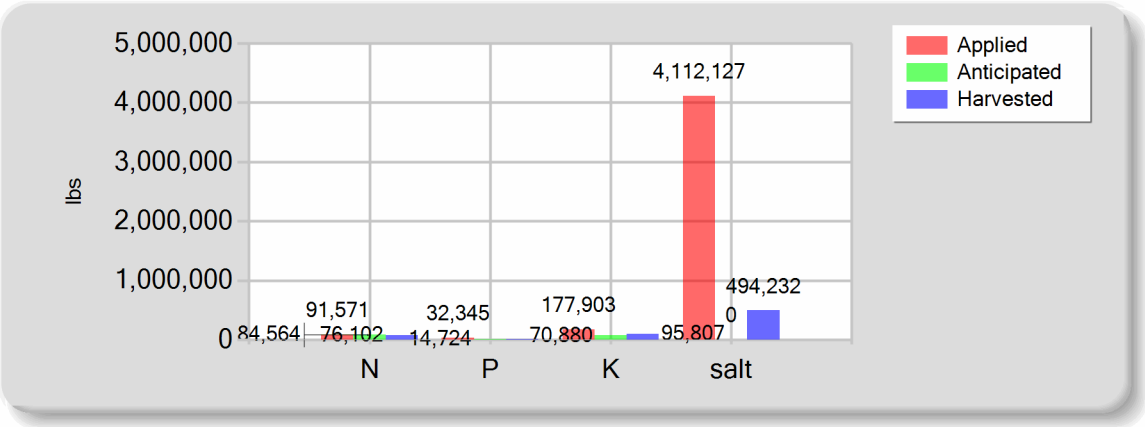
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

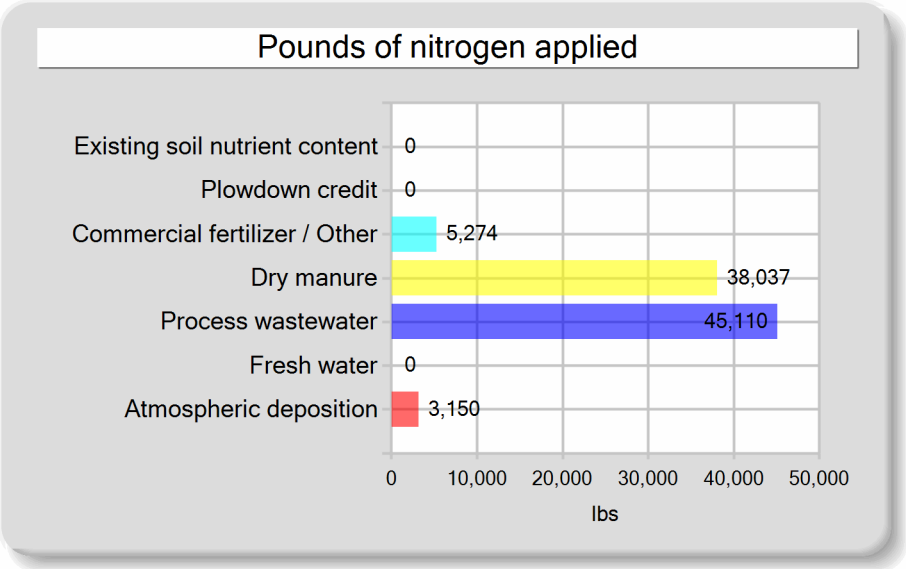
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

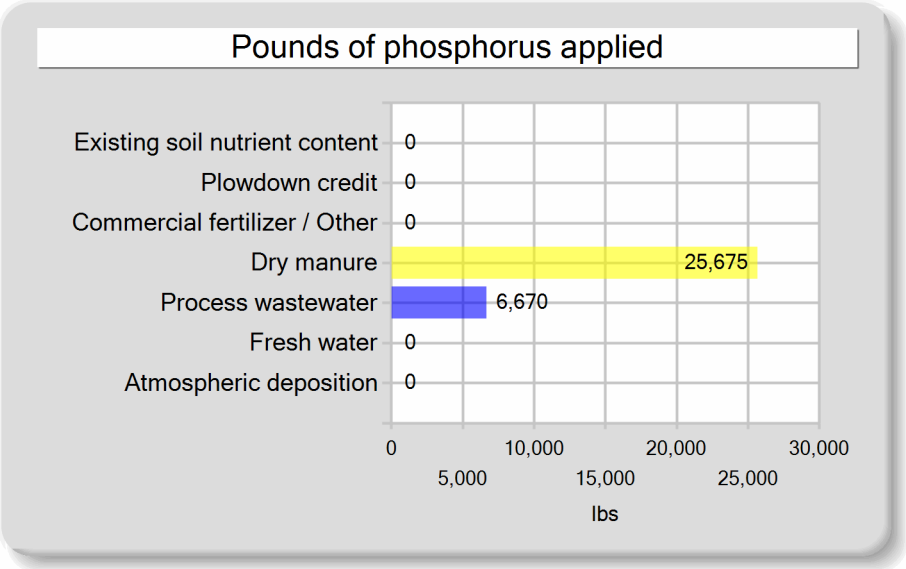
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	5,273.60	0.00	0.00	0.00
Dry manure	38,036.98	25,675.00	115,763.06	3,590,742.20
Process wastewater	45,110.17	6,670.16	62,139.45	433,987.23
Fresh water	0.00	0.00	0.00	87,397.69
Atmospheric deposition	3,150.00	0.00	0.00	0.00
Total nutrients applied	91,570.75	32,345.16	177,902.51	4,112,127.12
Anticipated crop nutrient removal	84,564.00	14,723.80	70,880.30	0.00
Actual crop nutrient removal	76,101.98	16,441.23	95,807.13	494,232.38
Nutrient balance	15,468.77	15,903.92	82,095.38	3,617,894.74
Applied to removed ratio	1.20	1.97	1.86	8.32

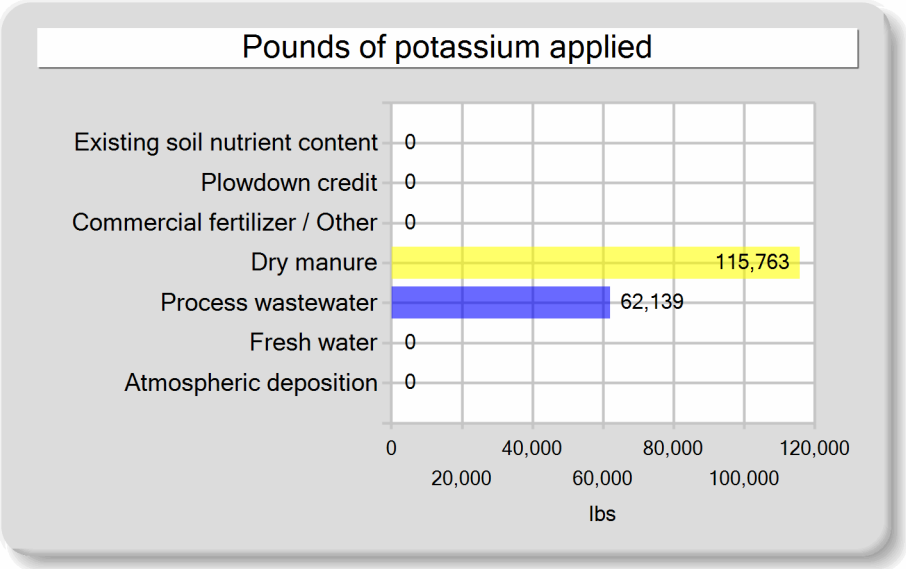
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

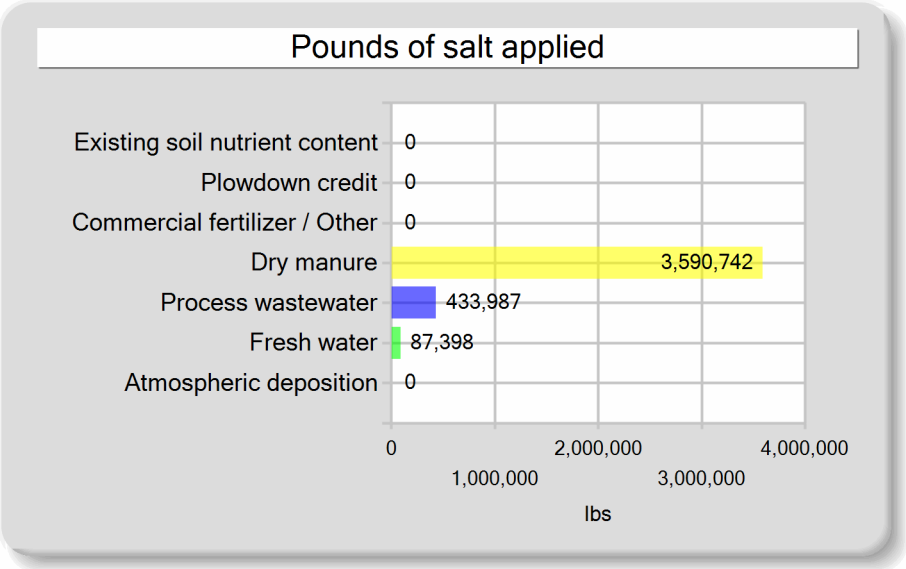


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Field 5 Oats (1.53) had a high nitrogen field ratio as crop failed to meet anticipated removal rate of 160 pounds of nitrogen/acre. The subject field removed only 125.9 pounds of nitrogen per acre. Yield for the subject field was adequate however a low nitrogen value from the plant tissue lab analysis resulted in 7.66 pounds of nitrogen removed per harvested ton compared the anticipated 10 pounds of nitrogen removed per harvested ton. Nitrogen application of 187 pounds in the form of solid manure was not considered excessive but in-line with that recommended in the dairy's nutrient management plan.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Johnny Silva (Apr 25, 2024 05:37 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Johnny Silva	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Apr 25, 2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
R & S Dairy
23243 Lone Tree Rd.
Escalon, CA 95320

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 2/13/24 2:16 PM
SUBMITTED BY: Jeanette
ANALYZED DATE: 2/14/24

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 21.2c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW 1	T4044025A	Electrical Conductivity (EC)	EPA 120.1 and	0.392	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	5.72	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	251	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	98.4	96.4 (T4044025A)	101 (T4044025A)	4.85
Electrical Conductivity (EC)	98.4			0.077
Nitrate Nitrogen (NO3-N)	102	102 (T4044004A)	102 (T4044004A)	0.352

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

Denele Analytical, Inc.

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Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
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Compliance Analysis Report

Customer
R & S Dairy
23243 Lone Tree Rd.
Escalon, CA 95320

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 2/13/24 2:16 PM
SUBMITTED BY: Jeanette
ANALYZED DATE: 2/14/24

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 18.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW 2	T4044025B	Electrical Conductivity (EC)	EPA 120.1 and	1.02	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	33.7	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	650	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	98.4	96.4 (T4044025A)	101 (T4044025A)	4.85
Electrical Conductivity (EC)	98.4			0.077
Nitrate Nitrogen (NO3-N)	102	102 (T4044004A)	102 (T4044004A)	0.352

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Compliance Analysis Report

Customer
R & S Dairy
23243 Lone Tree Rd.
Escalon, CA 95320

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 2/13/24 2:16 PM
SUBMITTED BY: Jeanette
ANALYZED DATE: 2/14/24

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 19.4c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW 3	T4044025C	Electrical Conductivity (EC)	EPA 120.1 and	0.197	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	2.12	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	126	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	98.4	96.4 (T4044025A)	101 (T4044025A)	4.85
Electrical Conductivity (EC)	98.4			0.077
Nitrate Nitrogen (NO3-N)	102	102 (T4044004A)	102 (T4044004A)	0.352

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

DENELE ANALYTICAL, INC.

865 S. KILROY ROAD
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Rush

Client: R&S
Mailing address:
(not facility address)
Phone#:

Authorized Copy Released To: F&R AG SERVICES
Sampled By: Jeanette
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
H1 EC*, NO3-N, NH4-N*
H6 EC, TN, TDS
H7 Combination of H1 & H6
H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
H3 EC*, NH4-N, TKN, TP, K, TDS
H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
H8 Combination of H3 &H5
H9 Combination of H4 &H5

TILE DRAIN
H2 EC*, NO3-N, TP, NH4-N*, TDS

*Required Field Measurement
____ (Check here if performed)

DISCHARGE WATER
E1 EC*, Temp*, pH*, NO3-N, NH3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
E2 EC,* Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
E3 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, TP, Total & Fecal Coliforms
E4 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
M1 % Moisture
M3 TN TP, K, % Moisture
M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
M7 Combination of M3 & M4

2023

PLANT TISSUE
F18 % Moisture, Ash, TN, P, K, DM and as received
F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
K3 Soluble Phosphorus
K4 1ft NO3, Organic Matter
K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
K7 Combination of K3 & K4
K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	2/13/24		DW 1	Domestic			✓				21.2
2	↓		↓ 2	↓			✓				18.6
3			3				✓				19.4
4											
5											
6											
7											
8											
9											
10											

Analysis

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	Logged In By:	LABORATORY NUMBER
	Jeanette Sawyer	F&R AG SERVICES				
2nd		DENELE ANALYTICAL, INC. (NO)	2/13/24 14:16			T4044025

(LABORATORY USE ONLY)

2023 Annual Report R&S Dairy

Final Audit Report 2024-04-25

Created:	2024-04-24
By:	Vince Furtado (vfurtado@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAAnaxWBagfaOhc2MpA93VIjH1eP8ChBNVY

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